

[Secure Exit](#)

CO3-AT3-Debugging and Optimization

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QUESTIONS

Q1

ATM Withdrawal – Exception Handling Debugging

10 marks

**Q2**

Hospital Registration – NullPointerException Debugging

10 marks

**Q3**

E-Commerce Product Search – Array Exception Debugging

10 marks



Q3 E-Commerce Product Search – Array Exception Debugging

10 marks

An online shopping application allows customers to select a product using a product number from 1 to 5.

The following program contains an array indexing error.

Task:

1. Identify the error.
2. Correct the product-number and array-index logic.
3. Handle invalid product numbers.
4. Handle non-numeric input.
5. Submit the COMPLETE corrected Java program.
6. The program must display "Search completed" after processing.



SIMATS Online Programming Lab

JAVA PROGRAMMING

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Railway Reservation – Synchronization Debugging



```
class ProductSearch {  
  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);
```

10 marks

5/5 answered

```
String[] products = {  
    "Laptop",  
    "Mobile",  
    "Tablet",  
    "Headphone",  
    "Keyboard"  
};  
  
System.out.print("Enter product number (1-5): ");  
int productNumber = sc.nextInt();  
  
System.out.println("Selected product: "  
    + products[productNumber]);  
  
System.out.println("Search completed");  
}  
}
```

Correct the program so that product numbers 1, 2, 3, 4 and 5 correctly access the corresponding products.

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.*;  
2 class Main{  
3     public static void main(String[] args){  
4         Scanner sc = new Scanner(System.in);  
5         String[] products = {"Laptop", "Mobile", "Tab
```

Test Input

1

```
6      try{
7          int productNumber=sc.nextInt();
8          if (productNumber<1||productNumber>5){
9              throw new IllegalArgumentException(
10             }
11             System.out.println("Selected product: ").
12         }
13         catch(InputMismatchException e){
14             System.out.println("Invalid numeric inp
15         }
16         catch(IllegalArgumentException e){
17             System.out.println(e.getMessage());
18         }
19         finally{
20             System.out.println("Search completed");
21         }
22     }
23 }
```

Output

Selected product: Laptop
Search completed

⌚ Pending manual review

✓ Submitted